

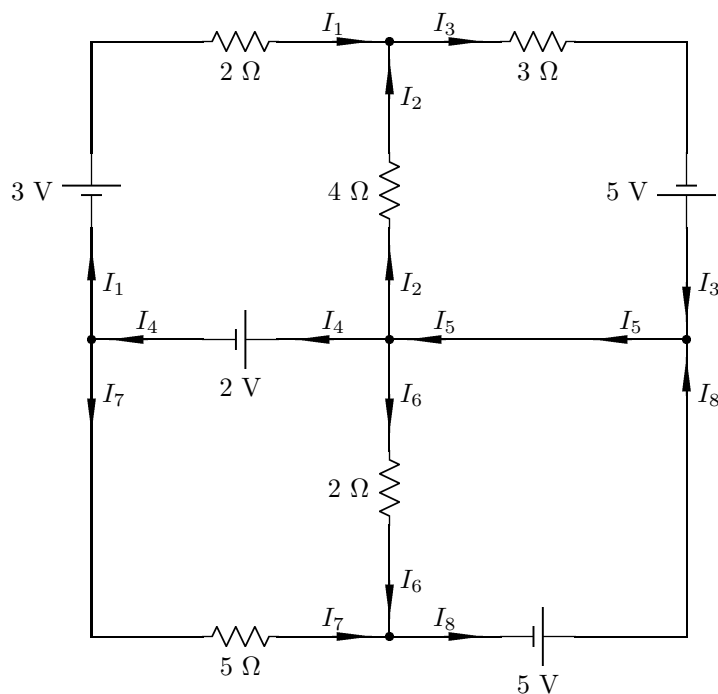
Extra Credit

Due Date: August 23, 2019 by 5:00pm

MAT 022A Linear Algebra

Instructor: Mikhail Gaerlan

Instructions: This assignment is worth 15 extra points added to Midterm 1. The figure below is an electrical circuit diagram. Attached is the section on electrical circuits from *Introductory Linear Algebra: An Applied First Course*. Use Kirchoff's Circuit Laws to set up a system of equations for the currents $I_1, \dots, I_8$ in each wire. Then use Gaussian Elimination to solve the system. Make sure to write down each row operation used.



- (c) Select the Arrow object and use the MATLAB matrix

$$\begin{bmatrix} \cos(\pi/4) & \sin(\pi/4) \\ -\sin(\pi/4) & \cos(\pi/4) \end{bmatrix}.$$

Describe the resulting image. What angle does it make with the positive horizontal axis? To help answer this question, use the grid button and then inspect the grid generated on the mapped arrow.

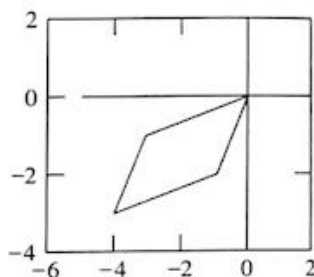
- (d) Using part (c), determine the coordinates of the top end of the arrow.

In Exercises ML.5 through ML.6, use the routine **planelt**. Matrix transformations from R^2 to R^2 are sometimes called plane linear transformations. The MATLAB routine **planelt** lets us experiment with such transformations by choosing the geometric operation we want to perform on a figure. The routine then uses the appropriate matrix to compute the image, displays the matrix, and keeps a graphical record of the original, the previous image, and the current image. Routine **planelt** is quite versatile since you can enter your own figure and/or matrix. To start this routine type **planelt**.

- ML.5.** Start routine **planelt**. Read the descriptions that appear and follow the on-screen directions until you get to FIGURE CHOICES. There select the triangle, choose to 'See the Triangle,' and then the option 'Use this Figure. Go to select transformations.'
- (a) Select the rotation and use a 45° angle. After the figures are displayed, press Enter. You will see the Plane Linear Transformation menu of options again. If you choose a transformation at this point, it will be used 'compositely' with the transformation just performed. Try this by choosing to reflect the current figure about the y-axis. The figures displayed show the original triangle, this triangle rotated through 45° , and then this image reflected about the y-axis. Record a sketch of the composite figure.
- (b) Reverse the order of the transformations in part (a). Record a sketch of the figure obtained from this composition. Compare this sketch with that

from part (a). If A is the matrix of the 45° rotation and B is the matrix of the reflection about the y-axis, then explain how we know that $BA \neq AB$.

- ML.6.** Use **planelt** with the parallelogram. Choose a composition of transformations so that the final figure is that shown in the accompanying figure.



- ML.7.** Orthogonal projections will play a fundamental role in a variety of situations for us later. While this section developed the algebra for computing projections, MATLAB can help present the geometric aspects. In MATLAB type **help project** and read the description. To start this routine, type **project** and then choose the demo for a first look at this routine. Use **project** to determine $\text{proj}_w \mathbf{u}$; that is, the projection of $\mathbf{u}$ onto $\mathbf{w}$, for each of the following pairs of vectors. Determine whether the projection is longer than the vector $\mathbf{w}$ and whether it is in the opposite direction.

(a) $\mathbf{u} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}, \mathbf{w} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

(b) $\mathbf{u} = \begin{bmatrix} 1 \\ -4 \end{bmatrix}, \mathbf{w} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$

(c) $\mathbf{u} = \begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix}, \mathbf{w} = \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix}$

(d) $\mathbf{u} = \begin{bmatrix} 4 \\ 6 \\ 0 \end{bmatrix}, \mathbf{w} = \begin{bmatrix} 2 \\ 3 \\ 8 \end{bmatrix}$

2.4 ELECTRICAL CIRCUITS

Prerequisite. Section 1.6, Solutions of Linear Systems of Equations.

In this section we introduce the basic laws of electrical-circuit analysis and then use these to analyze electrical circuits consisting of batteries, resistors, and wires.

A **battery** is a source of direct current (or voltage) in the circuit, a **resistor** is a device, such as a lightbulb, that reduces the current in a circuit by converting electrical energy into thermal energy, and a **wire** is a conductor that allows a free flow of electrical current. A simple electrical circuit is a closed connection of resistors, batteries, and wires. When circuits are represented by

diagrams, batteries, resistors, and wires are depicted as follows:

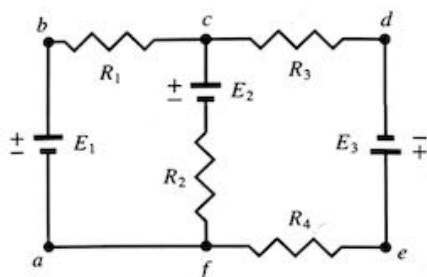
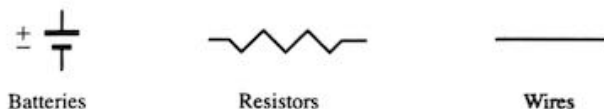


Figure 2.20 ▲

Figure 2.20 shows a simple electrical circuit consisting of three batteries and four resistors connected by wires.

The physical quantities used when discussing electrical circuits are current, resistance, and electrical potential difference across a battery. Electrical potential difference is denoted by E and is measured in volts (V). Current is denoted by I and is measured in amperes (A). Resistance is denoted by R and is measured in ohms (Ω). These units are all related by the equation

$$\text{One Volt} = (\text{One Ampere}) \times (\text{One Ohm}).$$

The electrical potential difference of a battery is taken as positive when measured from the negative terminal (–) to the positive terminal (+), and negative when measured from the positive terminal (+) to the negative terminal (–). The electrical potential difference across a resistor (denoted by V) depends on the current flowing through the resistor and its resistance, and is given by Ohm's law:

$$V = \pm IR.$$

The negative (–) sign is used when the difference is measured across the resistor in the direction of the current flow, while the positive (+) sign is used when the difference is measured across the resistor in the direction opposite the current flow.

All electrical circuits consist of voltage loops and current nodes. A **voltage loop** is a closed connection within the circuit. For example, Figure 2.20 contains the three voltage loops

$$a \rightarrow b \rightarrow c \rightarrow f \rightarrow a,$$

$$c \rightarrow d \rightarrow e \rightarrow f \rightarrow c,$$

and

$$a \rightarrow b \rightarrow c \rightarrow d \rightarrow e \rightarrow f \rightarrow a.$$

A **current node** is a point where three or more segments of wire meet. For example, Figure 2.20 contains two current nodes at points

$$c \text{ and } f.$$

Points a , b , d , and e are not current nodes since only two wire segments meet at these points.

The physical laws that govern the flow of current in an electrical circuit are conservation of energy and conservation of charge.

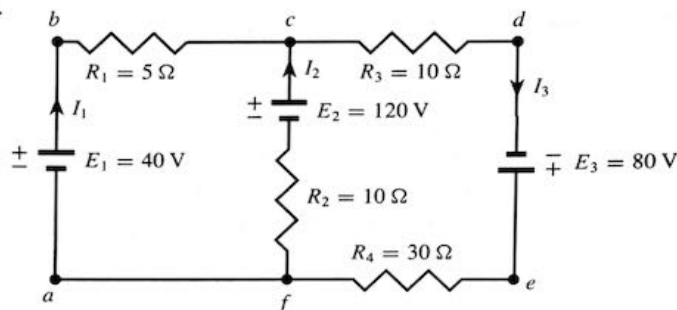
- The *conservation of energy* appears in what is known as **Kirchhoff's voltage law**: Around any voltage loop, the total electrical potential difference is zero.
- The *conservation of charge* appears in what is known as **Kirchhoff's current law**: At any current node, the flow of all currents into the node equals the flow of all currents out of the node. This guarantees that no charge builds up or is depleted at a node, so the current flow is steady through the node.

We are now ready to apply these ideas and the methods for solving linear systems to solving problems involving electrical circuits, which have the following general format. In a circuit containing batteries, resistors, and wires, determine all *unknown* values of electrical potential difference across the batteries, resistance in the resistors, and currents flowing through the wires, given enough *known* values of these same quantities. The following example illustrates this for the standard situation when the unknowns are the currents.

EXAMPLE 1

Figure 2.21 shows the circuit from Figure 2.20 with the batteries having the indicated electrical potentials, measured from the negative terminal to the positive one, and the resistors having the indicated resistances. The problem is to determine the currents that flow through each segment of the circuit.

Figure 2.21 ►



We begin by assigning currents to each segment of the circuit that begins at some node point and ends at some other node point (with no other node points in between). For example, in Figure 2.21, we assign I_1 to the segment $f \rightarrow a \rightarrow b \rightarrow c$, I_2 to the segment $f \rightarrow c$, and I_3 to the segment $c \rightarrow d \rightarrow e \rightarrow f$. In addition, we arbitrarily assign directions to these currents as indicated by the arrows in Figure 2.21. If the assigned direction is correct, the computed value of the current will be positive, while if the assigned direction is incorrect, the computed value of the current will be negative. The latter situation then indicates that the actual direction of current flow is just opposite to the assigned one. Using Kirchhoff's current law (the sum of the incoming currents = the sum of the outgoing currents) at points c and f , we have

$$I_1 + I_2 = I_3$$

and

$$I_3 = I_1 + I_2, \quad (1)$$

respectively. Since these two equations contain the same information, only one of them is needed. In general, if a circuit has n nodes, Kirchhoff's current law will yield $n - 1$ useful equations and one equation that is a linear combination of the other $n - 1$ equations.

Next we use Kirchhoff's voltage law. Starting at point a and moving through the battery from $(-)$ to $(+)$ to point b , the potential is $+40$ V. Moving from point b to point c through the $5\text{-}\Omega$ resistor results in a potential difference of $-5I_1$. Moving from point c to point f through the 120-V battery and $10\text{-}\Omega$ resistor results in a potential difference of -120 V (across the battery) and a potential difference of $+10I_2$ (across the resistor). Finally, moving along the wire from point f to point a results in no potential difference. To summarize,

applying Kirchhoff's voltage law around the closed loop $a \rightarrow b \rightarrow c \rightarrow f \rightarrow a$ leads to

$$(+E_1) + (-R_1 I_1) + (-E_2) + (+R_2 I_2) = 0$$

or

$$(+40) + (-5I_1) + (-120) + (10I_2) = 0,$$

which simplifies to

$$I_1 - 2I_2 = -16. \quad (2)$$

In a similar way, applying Kirchhoff's voltage law around the closed loop $c \rightarrow d \rightarrow e \rightarrow f \rightarrow c$ leads to

$$(-R_3 I_3) + (+E_3) + (-R_4 I_3) + (-R_2 I_2) + (+E_2) = 0$$

or

$$(-20I_3) + (+80) + (-30I_3) + (-10I_2) + (+120) = 0.$$

This simplifies to $10I_2 + 50I_3 = 200$, or

$$I_2 + 5I_3 = 20. \quad (3)$$

Note that the equation resulting from the voltage loop, $a \rightarrow b \rightarrow c \rightarrow d \rightarrow e \rightarrow f \rightarrow a$, becomes

$$(+E_1) + (-R_1 I_1) + (-R_3 I_3) + (+E_3) + (-R_4 I_3) = 0$$

or

$$(+40) + (-5I_1) + (-20I_3) + (+80) + (-30I_3) = 0,$$

which simplifies to

$$I_1 + 10I_3 = 24.$$

But this is just the linear combination Equation (2) + 2 Equation (3), and it therefore provides no new information. Thus this equation is redundant and can be omitted. In general, a larger outer loop like $a \rightarrow b \rightarrow c \rightarrow d \rightarrow e \rightarrow f \rightarrow a$ provides no new information if all its inner loops, like $a \rightarrow b \rightarrow c \rightarrow f \rightarrow a$ and $c \rightarrow d \rightarrow e \rightarrow f \rightarrow c$, have already been included.

Equations (1), (2), and (3) lead to the linear system

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 0 \\ 0 & 1 & 5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -16 \\ 20 \end{bmatrix}.$$

Solving this for I_1 , I_2 , and I_3 yields (verify)

$$I_1 = -3.5 \text{ A}, \quad I_2 = 6.25 \text{ A}, \quad \text{and} \quad I_3 = 2.75 \text{ A}.$$

The *negative* value of I_1 indicates that its true direction is opposite that assigned in Figure 2.21. ■

In general, for an electrical circuit consisting of batteries, resistors, and wires and having n different current assignments, Kirchhoff's voltage and current laws will always lead to n linear equations that have a unique solution.

Key Terms

Battery
Resistor
Wire
Voltage loop

Current node
Kirchhoff's voltage law
Kirchhoff's current law